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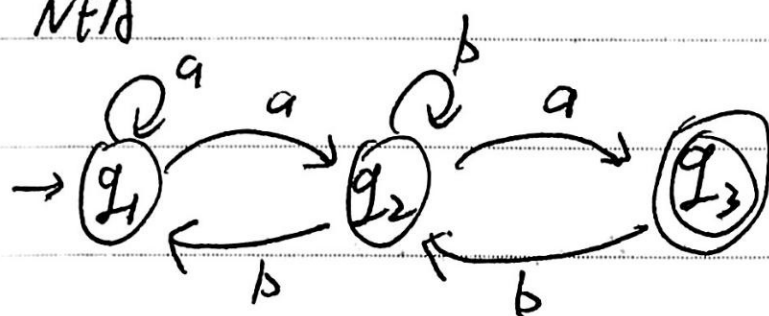
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(50) Designing Regular Expression

Find the Regular Expression for the following

NFA



$\{a, b\}$

$$q_3 = q_2 a \rightarrow (2)$$

$$q_2 = q_1 a + q_2 b + q_3 b \rightarrow (2)$$

$$q_1 = \epsilon + q_1 a + q_2 b \rightarrow (3)$$

$$(1) \rightarrow q_3 = q_2 a = (q_1 a + q_2 b + q_3 b) a$$

$$= q_1 a a + q_2 b a + q_3 b a \rightarrow (4)$$

$$(2) \rightarrow q_2 = q_1 a + q_2 b + q_3 b \quad \text{Putting value of } q_3 \text{ from (4)}$$
$$= q_1 a + q_2 b + q_2 a b$$

$$= q_1 a + q_2 (b + ab) \quad (5)$$

$$R = Q + R P \Rightarrow \text{use Arden's Theorem}$$

$$= Q P^*$$

$$q_2 = q_1 a (b + ab)^*$$



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$$\begin{aligned} \textcircled{3} \quad q_1 &= \epsilon + q_1 a + q_2 b \rightarrow \text{substitute with } \textcircled{5} \\ &= \epsilon + q_1 a + (q_1 a)(b+ab)^* \cdot b \\ &= \epsilon + q_1 (a + a(b+ab)^*) b \\ &= a + R P \\ &= \epsilon \cdot ((a + a(b+ab)^*) b)^* \\ &= ((a + a(b+ab)^*) b)^* \rightarrow \textcircled{6} \end{aligned}$$

Final state

$$\textcircled{q_3} = q_2 a$$

$$\begin{aligned} &= \cancel{q_1 a} + \cancel{q_2} \quad q_1 a(b+ab)^* a \\ &= (a + a(b+ab)^* b)^* a(b+ab)^* a \end{aligned}$$

$\Rightarrow$ regular expression for NFA

$$= (a(b+ab)^* b)^* a(b+ab)^* a$$

$$= (a(b+ab)^* b)^* a(b+ab)^* a$$

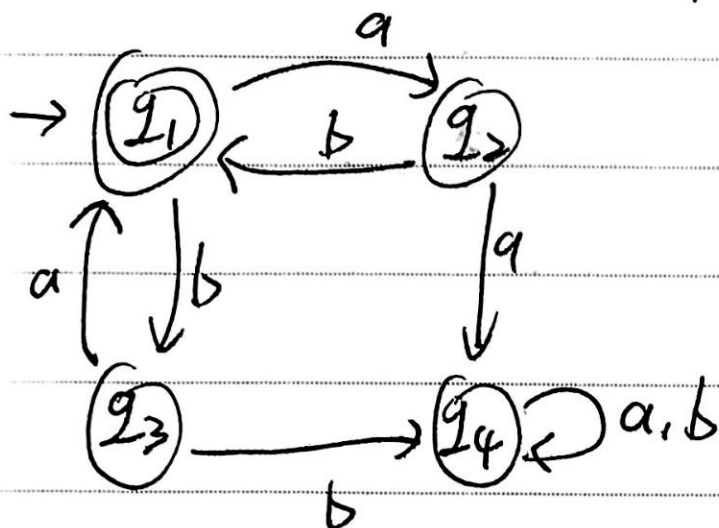


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(51) Design Regular Expression - Examples



$$① \quad q_1 = \epsilon + q_2 b + q_3 a$$

$$② \quad q_2 = q_1 a \rightarrow$$

$$③ \quad q_3 = q_1 b$$

$$④ \quad q_4 = q_2 a + q_3 b + q_4 (a+b)$$

$$Q \Rightarrow q_1 = \epsilon + q_2 b + q_3 a$$

$$= \epsilon + q_1 a b + q_1 b a$$

"Arden's Theorem"

$$= \epsilon + q_1 (ab + ba) = \epsilon + \text{RP}$$

$$= \epsilon \cdot (ab + ba)^* = \epsilon (ab + ba)^* \checkmark$$

$$y = e^{\int p(x) dx} \cdot \left(C + \int Q(x) e^{-\int p(x) dx} dx \right)$$

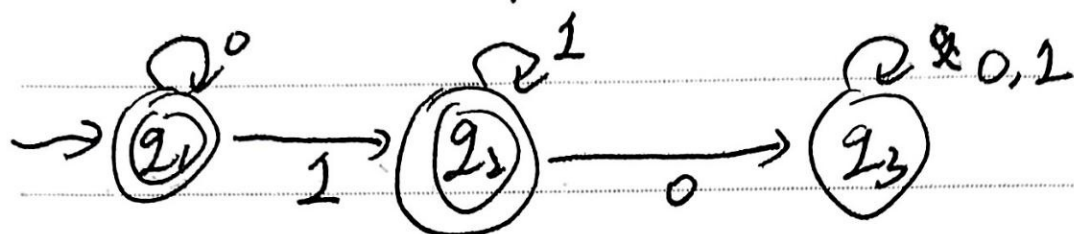


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(52) Design Regular Expression from DFA EX
(with multiple final states)



$$① \quad q_1 = q_1 0 + \epsilon$$

$$② \quad q_2 = q_1 1 + q_2 1$$

$$③ \quad q_3 = q_2 0 + q_3 0 + q_3 1$$

$$\Rightarrow q_1 = \epsilon + q_1 0 = \epsilon 0^* = 0^*$$

$Q + RP$

$$q_2 = q_1 1 + q_2 1 = 0^* 1 + q_2 1$$

$$= Q + RP$$

$$= 0^* 1 \cdot 1^* \quad \text{~~0^* 1^*~~}$$

$$\text{final states: } q_1 + q_2 = 0^* + 0^* 1^*$$

$$= \text{~~0^* 1^*~~}$$

$$\epsilon + R R^* = R^* \rightarrow = 0^* (\epsilon + 1 1^*)$$

$$= \text{~~1 1^*~~}$$

$$= 0^* 1^* = 0^* 1^*$$